MKALAMA DISTRICT COUNCIL
FORM ONE TERMINAL EXAMINATION
PHYSICS

Time: 2:30 Hours **Year 2025**

Instructions

1. This paper consists of sections A, B and C with a total of **ten (10)** questions.
2. Answer **all** questions
3. Where necessary the following constant may be used:
   (i) Gravitational field strength of the earth, **g = 10N/Kg**

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SECTION A (15 Marks)

Answer **al**l questions in this section.

1. For each of the items (i) – (x) choose the correct answer from the given alternatives and write its letter in your answer sheet provided.

(i.) A person who deals with the study of physics is called:-
A. Physikos B. Physicist
C. Scientists D. philosophers

(ii.) An empty glass cup was placed on a digital balance and its mass was **43.63g**. water was then added into the cup and the balance recorded a new mass of **71.06g** what was the exact mass of the water added into the cup:-
A. 114.69g B. 27.43g
C. 71.06g D. 43.63g

(iii.) Which of the following questions need to be explained by physics?
A. How water corrode iron? B. Why the person has to maintain balanced diet?
C. How plants synthesize food from sunlight? D. How rainbow is formed?

(iv.) The branch of physics that deals with the study of universe(cosmos) and everything contained in it is known as
A. Astronomy B Geophysics C Electronics D Atomic physics

(v.) Which of the following apparatus is used to measure the volume of an irregular solid?
A. pipette B. Burette
C. Measuring cylinder. D. Meter rule

(vi.) Which of the following is a set of equipment used in school that use the knowledge of physics.
A. Sawmills, cranes and spanners B fax machines, air planes and train
C Computers, printers and projectors D X-ray, ultrasound and syringe

(vii.) Physics is the study which deals with matter. What does it relate to?
A Power B Energy C Force D Work

(viii.) Joseph a form one student at Ilunda secondary school measures a cuboid and obtain the following dimensions length 3cm, width 4cm, and height 10cm. What was the actual volume obtained?
A. $120cm^3$ B. 240ml
C. $120m^3$ C. $240m^3$

(ix.) Which of the following scientific statements needs to be proved through scientific research?
A Hypothesis B Principle C Proposal D Conclusion

(x.) A set of techniques used by scientist to investigate a problem refers to:-
A. Data interpretation B. Scientific method
C. Data presentation D. Measurement.

2. Match the uses of instrument in **List A** with a correct name of the instrument in **List B** by writing a letter of the correct response in your answer sheet provided.

List A	List B
i. Ruler ii. Tape measure iii. Vernier caliper iv. Micrometer screw gauge v. Stopwatch.	A. It measure diameter of thin objects with accuracy of 0.01mm B. It measure the weight of an object C. It measure the distance or length of object D. It measure small length of an object with accuracy of 0.01cm E. Measure time F. It measure the curved surface, large distance and irregular shape.

SECTION B (70 Marks)

Answer **all** questions in this section.

3. (a) Why physics is important to our everyday life? Give four reason to support your answer **(8 marks)**
   (b) Mention four (4) vessels that rely on the laws of physics. **(2 marks)**
4. (a) Outline three (3) basic fundamental physical quantities and SI Unit. **(6 marks)**
   (b) Mention four (4) instruments used to measure length. **(4 marks)**
5. (a) Differentiate between parallax error and zero error. **(2 marks)**
   (b) Mention two instruments that are used to measure volume of irregular object. **(4 marks)**
   (c) Asha a form one student in Ilunda secondary school obtained the following results from her experiment.
   - Mass of empty beaker = 48g
   - Mass of beaker + water = 60g

   Calculate the mass of water only. **(4 marks)**
6. (a) Explain how each of the following quantities differ from each other
   (i) Density and relative density. **(2 marks)**
   (ii) Mass and weight. **(2 marks)**
   (b) The water collected with a cylinder during an experiment using a eureka can was $30cm^3$. When the object that displaced this water was dried and weighed its mass was found to be 90g, calculate its density. **(6 marks)**
7. (a) Differentiate between vector quantities and scalar quantities. **(5 marks)**
   (b) Mention four (4) areas where physics is applied. **(5 marks)**
8. (a) What is scientific method? **(3 marks)**
   (b) Form one student at Ilunda secondary school were asked to find out whether girls in their class perform better than boys in physics. Suggest seven (7) steps in chronological order to be followed in their experiment. **(7 marks)**
9. (a) Define the term **Force** and state its SI unit **(3 marks)**
   (b) (i) If an object weight 30N on the earth what it's mass? **(3 marks)**
   (ii) Differentiate between contact force and non-contact force **(4 marks)**

SECTION C (15 Marks).

Answer question **ten (10)**

10. A rectangular block measures height 1.00cm, width 2.50cm, and length 4.00cm
    (a) What instrument was used to measure the sides of the of the rectangular block **(3 marks)**
    (b) Calculate the volume of the rectangular block. **(6 marks)**
    (c) If the mass of the rectangular block is 20g calculate the density of the block. **(6 marks)**